GreenLit Go - Movie Success Prediction Using Data Analytics

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Abstract

Producing a movie is an incredibly expensive gamble. Despite the hundreds of millions of dollars on the line, studios still largely rely on gut instinct and past experience when predicting if a project will succeed or fail computationally. "GreenLit GO" was designed to address this problem directly. We built a practical, web-based platform that injects data science into the theatrical greenlighting process.

To make this work, we sourced data on over 2,180 films across multiple industries—including Hollywood and Bollywood—pulling from TMDB, OMDB, YouTube, and Box Office Mojo. We processed this raw information into 46 engineered features that capture the actual, real-world drivers of box office momentum. These metrics track the current popularity of the cast and director, the online hype generated by the trailer, seasonal release patterns, and relative budget sizes.

For the predictive engine itself, we implemented a Stacking Ensemble architecture. By layering algorithms like Random Forest, XGBoost, and LightGBM—and fine-tuning them via Optuna's Bayesian optimization—the model achieves an R^2 score of 0.75. It categorizes potential films into five outcomes ranging from 'Flop' to 'Blockbuster'. Rather than operating as a black box, the system uses SHAP to provide explainable AI, breaking down exactly which factors influenced every single prediction.

The full-stack application brings all of this to life. Built with a React and Vite frontend, a Node.js backend, and a Python Flask microservice for the ML pipeline mapping to MongoDB Atlas, it acts as a complete studio toolkit. Beyond simple number-crunching, users can explore industry trends through interactive dashboards, test out different production variables in a "what-if" sandbox, and even use the Gemini API to analyze raw screenplay PDFs for immediate narrative feedback.